作业 1

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2025 年 9 月 23 日

问题 (1). 对 $n>1$ 时，$H(x)=\begin{cases}1, & x_i>0, i=1,2,\ldots,n\\ 0, & \text{if not}\end{cases}$，则：

$$\frac{\partial^n H(x)}{\partial x_1\ldots\partial x_n}=\delta(x)$$

Proof. 我们用 $h(x)$ 来表示一维的 Heaviside 函数，则

$$H(x)=\begin{cases}1, & x_i>0, i=1,2,\ldots,n\\ 0, & \text{if not}\end{cases}=h(x_1)h(x_2)\ldots h(x_n)$$

同时，对于一维的 Heaviside 函数我们有

$$\frac{d}{dx_i}h(x_i)=\delta(x_i)$$

故

$$\frac{\partial^n H(x)}{\partial x_1\ldots\partial x_n}=\delta(x_1)\delta(x_2)\cdots\delta(x_n)=\delta(x)$$

事实上，对 $\forall\varphi\in C_0^\infty(\mathbb{R}^n)$，广义函数的微分运算，我们有：

$$\left\langle\frac{\partial^n H}{\partial x_1\cdots\partial x_n},\varphi\right\rangle=(-1)^n\left\langle H,\frac{\partial^n\varphi}{\partial x_1\cdots\partial x_n}\right\rangle=(-1)^n\int_{[0,\infty)^n}\frac{\partial^n\varphi}{\partial x_1\cdots\partial x_n}dx=\varphi(0)=\langle\delta,\varphi\rangle.$$

□

问题 (2). 设 $a(x)\in C^\infty(\Omega), u\in\mathscr{D}'(\Omega)$，证明：

$$\frac{\partial}{\partial x_i}(au)=a\frac{\partial u}{\partial x_i}+\frac{\partial a}{\partial x_i}u$$

Proof. 对 $\forall\varphi\in C_0^\infty(\Omega)$，由广义函数的微分运算，我们有：

$$\left\langle\frac{\partial}{\partial x_i}(au),\varphi\right\rangle=-\left\langle au,\frac{\partial\varphi}{\partial x_i}\right\rangle$$

由广义函数的乘子运算，我们有：

$$\left\langle au,\frac{\partial\varphi}{\partial x_i}\right\rangle=\left\langle u,a\frac{\partial\varphi}{\partial x_i}\right\rangle$$

由于 $a, \varphi \in C^\infty(\Omega)$，故由微分法则可知：

$$\frac{\partial(a\varphi)}{\partial x_i} = a\frac{\partial \varphi}{\partial x_i} + \varphi\frac{\partial a}{\partial x_i}$$

故我们有：

$$\left\langle u, a\frac{\partial \varphi}{\partial x_i}\right\rangle = \left\langle u, \frac{\partial(a\varphi)}{\partial x_i} - \varphi\frac{\partial a}{\partial x_i}\right\rangle = \left\langle u, \frac{\partial(a\varphi)}{\partial x_i}\right\rangle - \left\langle u, \varphi\frac{\partial a}{\partial x_i}\right\rangle = -\left\langle \frac{\partial u}{\partial x_i}, a\varphi\right\rangle - \left\langle u, \varphi\frac{\partial a}{\partial x_i}\right\rangle$$

我们将其带回最初式子有：

$$\left\langle \frac{\partial}{\partial x_i}(au), \varphi\right\rangle = -\left\langle au, \frac{\partial \varphi}{\partial x_i}\right\rangle = \left\langle \frac{\partial u}{\partial x_i}, a\varphi\right\rangle + \left\langle u, \varphi\frac{\partial a}{\partial x_i}\right\rangle = \left\langle a\frac{\partial u}{\partial x_i}, \varphi\right\rangle + \left\langle \frac{\partial a}{\partial x_i}u, \varphi\right\rangle = \left\langle a\frac{\partial u}{\partial x_i} + \frac{\partial a}{\partial x_i}u, \varphi\right\rangle$$

故由广义相等我们可知：

$$\frac{\partial}{\partial x_i}(au) = a\frac{\partial u}{\partial x_i} + \frac{\partial a}{\partial x_i}u$$

□

问题 (3). 若广义函数 u 适合 $\check{u} = u$（$\check{u} = -u$），则称 u 为偶（奇）广义函数。证明：

(1) $\delta(x)$ 与常值函数 c 皆为偶广义函数。

(2) 偶广义函数之微商是奇广义函数。

(3) 任一广义函数 $u \in \mathscr{D}'(\Omega)$ 皆可唯一地分解为一个偶广义函数与一个奇广义函数之和如下：

$$u(x) = \frac{1}{2}(u(x) + \tilde{u}(x)) + \frac{1}{2}(u(x) - \tilde{u}(x)).$$

(4) $u \in \mathscr{D}'(\Omega)$ 为偶广义函数当且仅当对任一奇的 $\varphi \in \mathscr{D}(\Omega)$ 均有 $\langle u, \varphi\rangle = 0$。

Proof. 对 $\forall \varphi \in C_0^\infty(\Omega)$：

(1) ①我们有 $\langle \delta, \varphi\rangle = \varphi(0)$，$\langle \check{\delta}, \varphi\rangle = \langle \delta, \check{\varphi}\rangle = \check{\varphi}(0) = \varphi(-0) = \varphi(0) = \langle \delta, \varphi\rangle \Rightarrow \delta(x) = \check{\delta}(x)$ 即 $\delta(x)$ 为偶广义函数；

②我们有 $\langle \check{c}, \varphi\rangle = \langle c, \check{\varphi}\rangle = c\int \check{\varphi}(x)dx = c\int \varphi(-x)dx = c\int \varphi(x)dx = \langle c, \varphi\rangle \Rightarrow c = \check{c}$ 即常值函数 c 为偶广义函数.

(2) 设 u 为偶广义函数满足 $u = \check{u}$，则：

$$\langle (\check{\frac{\partial u}{\partial x_i}}), \varphi\rangle = \langle \frac{\partial u}{\partial x_i}, \check{\varphi}\rangle = -\langle u, \frac{\partial \check{\varphi}}{\partial x_i}\rangle = -\langle u, \frac{\partial(-\varphi)}{\partial x_i}\rangle = \langle u, \frac{\partial \varphi}{\partial x_i}\rangle = -\langle (\frac{\partial u}{\partial x_i}), \varphi\rangle = \langle -(\frac{\partial u}{\partial x_i}), \varphi\rangle$$

故由广义相等我们可知：

$$(\check{\frac{\partial u}{\partial x_i}}) = -(\frac{\partial u}{\partial x_i})$$

即为奇广义函数.

(3) 令 $u_1 = \frac{1}{2}(u+\check{u})$，$u_2 = \frac{1}{2}(u-\check{u})$。其中：$\check{u_1} = \frac{1}{2}(\check{u}+u) = u_1$ 是偶广义函数；$u_2\check{u_2} = \frac{1}{2}(\check{u}-u) = -u_2$ 是奇广义函数。故：

$$u_1 + u_2 = \frac{1}{2}(u+\check{u}) + \frac{1}{2}(u-\check{u}) = u$$

下面我们来证明分解的**唯一性**：若 $u = u_1' + u_2'$，其中 u_1' 是偶广义函数，u_2' 是奇广义函数，则 $\check{u} = u_1' - u_2'$。联立 $u = u_1' + u_2'$ 与 $\check{u} = u_1' - u_2'$，解得 $u_1' = \frac{1}{2}(u+\check{u})$，$u_2' = \frac{1}{2}(u-\check{u})$，故分解唯一。

(4) ($\Rightarrow$) 若 u 是偶广义函数，即 $\check{u}=u$。对任一奇的 $\varphi\in\mathscr{D}(\Omega)$，有 $\check{\varphi}=-\varphi$。则 $\langle u,\varphi\rangle=\langle\check{u},\varphi\rangle=\langle u,\check{\varphi}\rangle=\langle u,-\varphi\rangle=-\langle u,\varphi\rangle$，故 $2\langle u,\varphi\rangle=0$，即 $\langle u,\varphi\rangle=0$。

($\Leftarrow$) 若对任一奇的 $\varphi\in\mathscr{D}(\Omega)$，$\langle u,\varphi\rangle=0$。令 $\varphi=\varphi_1+\varphi_2$（其中 φ_1 是偶函数，φ_2 是奇函数），则 $\langle u,\varphi\rangle=\langle u,\varphi_1\rangle+\langle u,\varphi_2\rangle=\langle u,\varphi_1\rangle$。同时 $\langle\check{u},\varphi\rangle=\langle u,\check{\varphi}\rangle=\langle u,\varphi_1\rangle$，故 $\langle u-\check{u},\varphi\rangle=0$ 对所有 $\varphi\in\mathscr{D}(\Omega)$ 成立，即 $u=\check{u}$，u 是偶广义函数。

□

问题 (4). 证明：若 $f(x)=H(x)\cos x$，$g(x)=H(x)\sin x$，则

$$f'(x)=\delta(x)-g(x),\quad g'(x)=f(x),$$

因此 g 与 f 分别适合微分方程

$$u''+u=\delta \quad 与 \quad u''+u=\delta'.$$

Proof. $f(x)=H(x)\cos x\Rightarrow f'(x)=H'(x)\cos x+H(x)(\cos x)'$。已知 $H'(x)=\delta(x)$，$(\cos x)'=-\sin x$，故 $f'(x)=\delta(x)\cos 0-H(x)\sin x=\delta(x)-g(x)$.

$g(x)=H(x)\sin x$，同理可得 $g'(x)=H'(x)\sin x+H(x)(\sin x)'=\delta(x)\sin 0+H(x)\cos x=f(x)$.

对 $g(x)$，求二阶导数 $g''(x)=f'(x)=\delta(x)-g(x)$，整理得 $g''+g=\delta$.

对 $f(x)$，求二阶导数 $f''(x)=(f')'=(\delta(x)-g(x))'=\delta'(x)-g'(x)=\delta'(x)-f(x)$，整理得 $f''+f=\delta'$.

因此 g 与 f 分别适合微分方程 $u''+u=\delta$ 与 $u''+u=\delta'$.

□